

IDIONG, ESANGUBONG JIMMY

ML/AI Engineering

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PROFESSIONAL SUMMARY

Full-Stack AI Engineer with a Computer Engineering background and end-to-end ownership of AI-powered applications across classical ML and LLM architectures. Experienced in building production-ready ML models and pipelines, from data ingestion and model training to API deployment and frontend integration. Applies MLOps fundamentals including experiment tracking, model registry, performance and drift monitoring. Interested in health AI.

SKILL

Backend & APIs

Python, FastApi, SQLAlchemy, PostgreSQL

Frontend:

TypeScript, Next.js, React, TailwindCss

Data & MLOps

Pandas, MLflow, Docker, AWS S3, GitHub Actions, Evidently, Airflow

Database & AI:

PostgreSQL, ChromaDB, FastEmbed, RAG, Google GenAI, Scikit-learn,

WORK EXPERIENCE

Xorion Network

Full-Stack Developer

June 2025 - Dec 2025

- Built and deployed the IDO website with full user functionality, enabling wallet connection and seamless participation in the token sale (Substrate + Ethereum).
- Engineered real-time portfolio dashboards, optimizing data flow between Substrate/Ethereum backends and the user interface.
- Architected cross-chain bridging functionality, contributing to the Relay implementation for Substrate-to-Ethereum token transfers.
- Implemented the release token feature to enable bridging from Ethereum back to Substrate, ensuring two-way token transfers
- Refactored legacy codebase, improving system maintainability and performance, while ensuring seamless wallet integration for high-volume token sales.

PROJECTS

DiaWatch - Diabetes Risk Prediction System

Python, Scikit-learn, CatBoost, Vercel, MLFlow, FastAPI, Docker, Google GenAI, TypeScript, PostgreSQL, HuggingFace

- Designed a multi-class diabetes risk classifier (Healthy/Prediabetes/Diabetes) on 19k+ NHANES records using CatBoost with custom clinical thresholds, SHAP and MICE-based imputation; tracked experiments via MLflow/DagsHub and monitored drift with Evidently.
- Built a Next.js + FastAPI full-stack platform with offline IndexedDB history, and a feedback loop for continuous model improvement; deployed frontend to Vercel and backend to HuggingFace Spaces docker sdk.
- Integrated Google GenAI to generate contextual, personalized health recommendations from SHAP feature attributions, translating model outputs into actionable lifestyle and screening guidance for users.

[Live](#) | [Code](#)

MediFact - Medical Retrieval-Augmented Generation (RAG) System

Python, FastAPI, ChromaDB, FastEmbed, LLaMA-3, AWS S3, Airflow, Docker, Modal, GitAction, HuggingFace

- Architected a decoupled, serverless RAG infrastructure using Modal for GPU-accelerated inference, reducing idle infrastructure costs by leveraging cold-start optimization. Strict prompting and citation guardrails.
- Engineered an automated MLOps pipeline using GitHub Actions, AWS S3(storing artifacts) and Airflow (development) for bi-monthly data ingestion, including hash-based change detection to minimize redundant processing of WHO datasets.
- Implemented a sophisticated retrieval strategy, including Semantic Chunking, Query Intent Classification (LLaMA-3), query expansion, token-aware context pruning and Cross-Encoder Re-ranking to ensure clinical accuracy.
- Optimized memory efficiency by deploying lightweight BGE-small embeddings via FastEmbed, achieving sub-second latency for semantic search in ChromaDB.

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EDUCATION

University of Uyo, B.Eng. Computer Engineering (GPA: 4.0/5.0)

October, 2023